

Integrazioni — API Reference (local-agent)

Riferimento rapido agli adapter in `src/integrations/`. Ogni modulo è ESM, senza dipendenze esterne oltre a Node built-in.

Indice

Modulo	Namespace	Config / Auth
Chrome CDP	<code>chromeCdp</code>	DevTools port
Google Auth	<code>googleAuth</code>	OAuth <code>~/.108ai/google-oauth.json</code>
Gmail	<code>gmail</code>	Token Google
Google Calendar	<code>googleCalendar</code>	Token Google
IMAP / PEC	<code>imapClient</code>	<code>~/.108ai/integrations/imap.json</code>
Outlook	<code>officeOutlook</code>	PowerShell + Outlook desktop
Excel / Word	<code>officeExcel</code> , <code>officeWord</code>	COM automation Windows
UI Automation	<code>uiAutomation</code>	Windows UIA
Vision LLM	<code>visionLlm</code>	Screenshot + gateway
Telegram	<code>telegramBot</code>	Bot token env
WhatsApp Business	<code>whatsappBusiness</code>	Meta Graph API config
WhatsApp Baileys	<code>whatsappBaileys</code>	Session locale
Fatture in Cloud	<code>fattureInCloud</code>	<code>FATTURE_IN_CLOUD_TOKEN</code> , <code>FATTURE_IN_CLOUD_COMPANY_ID</code>

Lazy loading

Il triage e altri hot path usano `loadIntegration(name)` da `lazy-loader.ts` per caricare i moduli solo al primo utilizzo.

```
import { loadIntegration } from '../integrations/lazy-loader.js';

const gmail = await loadIntegration<typeof import('../integrations/gmail.js')>('gmail');
const messages = await gmail.listMessages(token, { query: 'is:unread', maxResults: 10 });
```

Fatture in Cloud (Sprint 5 entry)

```
import * as fic from '../integrations/fatture-in-cloud.js';

if (fic.isFattureInCloudConfigured()) {
  const { invoices, total } = await fic.listOverdueInvoices(20);
}
```

Triage: con `billing.enabled: true` in `~/.108ai/triage.json`, `/triage` include fatture scadute come item source: `billing`.

Gmail — operazioni principali

Funzione	Descrizione
<code>listMessages(token, opts)</code>	Lista con query Gmail
<code>getMessage(token, id)</code>	Dettaglio singolo
<code>sendEmail(token, params)</code>	Invio / reply
<code>searchMessages(token, query)</code>	Wrapper search

MCP tools (extensions)

Tool referenziati come `mcp:<server>:<tool>` in agent/skill YAML. Richiedono server in `~/108ai/mcp.yml` con transport `stdio` o `sse` + url.

Estendere

- Creare `src/integrations/<nome>.ts` con tipi esportati e funzioni pure/async.
- Esportare namespace in `integrations/index.ts`.
- Aggiungere case in `lazy-loader.ts`.
- (Opzionale) Collegare al triage engine o a un command YAML `context.integration`.

Sicurezza

- Mai loggare token o PII.
- Timeout su ogni fetch esterno (15–30s).
- Secret solo in env o file cifrati (`key-vault.ts` per provider LLM).